Lab Assignment -2

Course- B. Tech
Course Code- CSET343
Year- 4^{th} Year

Type- Specialization Elective
Course Name- AI in Healthcare
Semester- VII

Objective: Perform data cleaning and enrichment on a sample clinical dataset with missing values.

Download the following dataset: [Heart Failure Dataset]

Perform all the following tasks on the above dataset.

1. Dataset Acquisition: Download Heart Failure Clinical Records Dataset
2. Load Dataset: Read dataset as pandas DataFrame, display first 5 rows.
3. Summarize Dataset: Report number of rows, columns, data types, missing values.
4. Visualize Missing Values: Create heatmap using seaborn to show missing values.
5. Plot Distributions: Generate histograms or boxplots for numerical features (e.g., age, ejection_fraction).
6. Impute Missing Values:
    - Numerical features: Use mean or median (e.g., serum_creatinine).
    - Categorical features: Use mode or "Unknown" (e.g., smoking).
7. Remove Outliers
8. Correct Inconsistencies: Check and fix invalid entries (e.g., negative age).
9. Feature Engineering:
    - Create age_group feature (bins: <40, 40–60, >60).
    - Create risk_score feature (e.g., normalized product of ejection_fraction, serum_creatinine).
10. Encode Categorical Variables: Apply one-hot or label encoding to categorical features (e.g., sex, smoking).
11. Normalize Features: Scale numerical features (e.g., age, platelets) to 0–1 using MinMaxScaler.
12. Compute Summary Statistics: Calculate mean, median, standard deviation before and after cleaning.
13. Validate Cleaning: Plot histograms or boxplots for cleaned numerical features.
14. Check Missing Values: Confirm no missing values remain or justify any remaining.